

Supplemental mathematical proofs

Appendix overview

This appendix contains the full mathematical details that are referenced in the main text. The appendix uses sections only, as requested. Each proof begins with a one sentence interpretation so that the formal derivation stays connected to the main insight.

Proof of the cluster-length proposition

Interpretation. This proof shows that the border score is a direct measure of expected local burst length.

Proof. Under the renewal cluster model, a cluster continues after each active occurrence with probability ρ_w and stops with probability $1 - \rho_w$. Hence

$$\mathbb{P}(C_w = k) = (1 - \rho_w)\rho_w^{k-1}, \quad k = 1, 2, \dots \quad (1)$$

which is the geometric distribution on $\{1, 2, \dots\}$. Therefore

$$\mathbb{E}[C_w] = \sum_{k=1}^{\infty} k(1 - \rho_w)\rho_w^{k-1} = \frac{1}{1 - \rho_w}. \quad (2)$$

Using the definition $\beta_w = \rho_w/(1 - \rho_w)$ yields

$$\frac{1}{1 - \rho_w} = 1 + \frac{\rho_w}{1 - \rho_w} = 1 + \beta_w. \quad (3)$$

This completes the proof. □

Proof of the variance proposition

Interpretation. This proof shows that valid self overlap creates positive local dependence and therefore larger variance of token counts.

Proof. For brevity write $I_t = I_t(w; X)$. Since $N_w = \sum_{t=1}^{T-m+1} I_t$, the variance decomposition gives

$$\text{Var}(N_w) = \sum_{t=1}^{T-m+1} \text{Var}(I_t) + 2 \sum_{1 \leq s < t \leq T-m+1} \text{Cov}(I_s, I_t), \quad (4)$$

which is the variance decomposition used in the main retained-information argument. Stationarity implies $\mathbb{P}(I_t = 1) = p_w$ and $\text{Var}(I_t) = p_w(1 - p_w)$.

Now fix a shift ℓ with $1 \leq \ell < m$. If $\ell \notin S(w)$, then two occurrences of w at positions t and $t + \ell$ are incompatible because the required prefix and suffix letters disagree. Hence

$$\mathbb{P}(I_t = 1, I_{t+\ell} = 1) = 0, \quad (5)$$

and

$$\text{Cov}(I_t, I_{t+\ell}) = -p_w^2. \quad (6)$$

If $\ell \in S(w)$, then the joint event is feasible and

$$\begin{aligned} \mathbb{P}(I_t = 1, I_{t+\ell} = 1) &= \mathbb{P}(I_t = 1) \mathbb{P}(I_{t+\ell} = 1 \mid I_t = 1) \\ &= p_w q_w(\ell). \end{aligned} \quad (7)$$

Therefore

$$\text{Cov}(I_t, I_{t+\ell}) = p_w q_w(\ell) - p_w^2. \quad (8)$$

This proves the local-covariance identity used in the main derivation. Summing over all pairs with the same shift gives

$$\begin{aligned} \text{Var}(N_w) &= (T - m + 1) p_w (1 - p_w) \\ &\quad + 2 \sum_{\ell=1}^{m-1} (T - m + 1 - \ell) \text{Cov}(I_t, I_{t+\ell}) \\ &\quad + R_{\text{long}}, \end{aligned} \quad (9)$$

where R_{long} collects shifts at least m . Dropping all nonpositive invalid overlap terms and all nonnegative contributions from R_{long} yields the lower bound stated in the main text. This completes the proof. $\square$

Proof of the retained-information theorem

Interpretation. This proof shows that a larger border score lowers retained information because suppression reacts to the larger burst scale of the response.

Proof. By definition,

$$I_w(\tau) = \mathbb{E}[\phi_\tau(a_w)^2 g_w^2]. \quad (10)$$

Since g_w is independent of a_w and $\mathbb{E}[g_w^2] < \infty$,

$$I_w(\tau) = \mathbb{E}[g_w^2] \mathbb{E}[\phi_\tau(a_w)^2]. \quad (11)$$

Likewise,

$$I_w(\infty) = \mathbb{E}[g_w^2] \mathbb{E}[\phi_\infty(a_w)^2]. \quad (12)$$

Because $\phi_\infty(u) = 1$ by definition of no suppression, we obtain

$$I_w(\infty) = \mathbb{E}[g_w^2]. \quad (13)$$

Hence

$$R_w(\tau) = \frac{I_w(\tau)}{I_w(\infty)} = \mathbb{E}[\phi_\tau(a_w)^2]. \quad (14)$$

Using the scale model $a_w \stackrel{d}{=} b_w Z$ yields

$$R_w(\tau) = \mathbb{E}[\phi_\tau(b_w Z)^2]. \quad (15)$$

This is the retained-information expectation identity used in the main text.

To prove monotonicity, let $b_1 \leq b_2$. Since $\phi_\tau(u)$ is nonincreasing in $|u|$ and $|b_1 Z| \leq |b_2 Z|$ pointwise whenever $|Z|$ is fixed, we have

$$\phi_\tau(b_1 Z)^2 \geq \phi_\tau(b_2 Z)^2 \quad \text{almost surely.} \quad (16)$$

Taking expectations gives

$$\mathbb{E}[\phi_\tau(b_1 Z)^2] \geq \mathbb{E}[\phi_\tau(b_2 Z)^2]. \quad (17)$$

Thus $R_w(\tau)$ is nonincreasing in b_w . Since $b_w = b_0(1 + \beta_w)$ with $b_0 > 0$, $R_w(\tau)$ is also nonincreasing in β_w . If ϕ_τ is strictly decreasing on a set of positive probability, then the inequality is strict for any $b_1 < b_2$ on a set of positive probability, which implies strict monotonicity. This completes the proof. $\square$

Proof of the compensation-weight theorem

Interpretation. This proof shows that the compensation weight should grow exactly where retained information becomes small.

Proof. The compensation objective is separable across words. For a fixed word $w \in \mathcal{H}$ define

$$J_w(\gamma_w) = \omega_w \frac{c_w}{\gamma_w^2 R_w(\tau)} + \lambda_\Gamma \gamma_w^2, \quad \gamma_w > 0. \quad (18)$$

Differentiating gives

$$J'_w(\gamma_w) = -2\omega_w \frac{c_w}{\gamma_w^3 R_w(\tau)} + 2\lambda_\Gamma \gamma_w. \quad (19)$$

Setting the derivative to zero yields

$$\lambda_\Gamma \gamma_w^4 = \frac{\omega_w c_w}{R_w(\tau)}. \quad (20)$$

Since $\gamma_w > 0$, the unique stationary point is

$$\gamma_w^\star = \left(\frac{\omega_w c_w}{\lambda_\Gamma R_w(\tau)} \right)^{1/4}. \quad (21)$$

The second derivative is

$$J''_w(\gamma_w) = 6\omega_w \frac{c_w}{\gamma_w^4 R_w(\tau)} + 2\lambda_\Gamma > 0, \quad (22)$$

so the stationary point is the unique global minimizer. Finally, because $\gamma_w^\star$ is proportional to $R_w(\tau)^{-1/4}$, any decrease of $R_w(\tau)$ with β_w implies a nondecreasing dependence of $\gamma_w^\star$ on β_w . If the decrease is strict, then the increase is strict. This completes the proof. $\square$

Additional derivation for Markov continuation mass

Interpretation. This derivation clarifies how the background Markov model enters the continuation mass and why the theory remains tied to genomic word structure.

For a valid shift $\ell \in S(w)$, the event $I_t(w) = 1$ fixes the letters $w_1, \dots, w_m$ on positions $t, \dots, t+m-1$. The event $I_{t+\ell}(w) = 1$ requires the suffix letters $w_{m-\ell+1}, \dots, w_m$ to be followed by the trailing block $w_{m-\ell+1}, \dots, w_m$ again. Under a first order Markov chain this gives

$$q_w(\ell) = \prod_{j=m-\ell+1}^{m-1} P_{w_j, w_{j+1}}, \quad (23)$$

up to the shared overlap block that is already fixed by $I_t(w) = 1$. Therefore the continuation mass can be estimated by

$$\hat{\rho}_w = \sum_{\ell \in S(w)} \prod_{j=m-\ell+1}^{m-1} \hat{P}_{w_j, w_{j+1}}, \quad (24)$$

where $\hat{P}$ is the empirical transition matrix estimated from the pretraining corpus. This appendix section is useful for implementation because it gives a direct route from tokenizer vocabulary to border scores without requiring labeled data. On heterogeneous genomes, $\hat{\rho}_w$ should be interpreted cautiously when w lies in repetitive or low-complexity regions because continuation mass can be inflated by tandem or interspersed repeats rather than by a single functional motif.

Implementation mapping from theory to trainer

Interpretation. This section specifies how tokenizer vocabulary, border scores, and compensation are instantiated in the released trainer, resolving the notational gap between vocabulary-aligned Γ and the deployed LoRA branch.

Tokenizer-to-word alignment. Each tokenizer token $w \in \mathcal{V}$ is mapped to a DNA string (BPE merge or k -mer in DNABERT-2). For token w of nucleotide length $m(w)$, the overlap set $S(w)$ and border polynomial $B_w(z)$ are computed on that string. The continuation mass $\hat{\rho}_w$ follows the Markov continuation formula above with $\hat{P}$ estimated from a background corpus. The border score is $\beta_w = \hat{\rho}_w / (1 - \hat{\rho}_w)$ when $\hat{\rho}_w < 1$.

Controlled metadata score. On synthetic border-aware splits, each sample stores a scalar `border_score` in CSV metadata aligned with the constructed difficulty of that sample. During training, the score is batch-normalized and passed to the compensation rule in the main text with λ_{comp} , c_{min} , and c_{max} from the YAML config.

External sequence-estimated score. When metadata is unavailable, a label-free score is computed from the input sequence (center-window JSD over k -mers), then train-quantile normalized to reduce saturation before compensation. The same compensation path is used; only the score source changes.

Low-rank branch. For each targeted linear layer W , LoRA factors A and B are attached as in standard LoRA. GERM-BO replaces the additive update $y = Wx + \text{scale} \cdot B(Ax)$ with $y = Wx + \text{scale} \cdot B(Ax \cdot c(x))$, where $c(x)$ is broadcast across valid token positions in a sequence. No separate Γ matrix is stored; the theoretical diagonal γ_w is realized through $c(x)$ derived from β_w or metadata.

Hyperparameter correspondence. Config field `compensation_strength` equals λ_{comp} in the main compensation rule. Fields `compensation_clip_min` and `compensation_clip_max` equal c_{min} and c_{max} . LoRA `rank` and `alpha` are r and the LoRA scaling numerator. The theoretical λ_{Γ} , τ , and α are not exposed as trainer knobs in the current release.